1. Geometric Distribution

- A geometric distribution is a discrete probability distribution that is modelled using a single parameter, p, which represents the probability of success.

1.1. Finding the probability using the distribution

- We say a continuous random variable, X, follows a geometric distribution $Geo(p)$. This is written as

$$X \sim Geo(p)$$

The probability of getting a success on the xth trial is found using

$$P(X = x) = p \times (1-p)^{x-1} \quad \text{[Given in MF19]}$$

Here

- x is a positive integer
  - $x \geq 1$
- $(1-p)$ denotes the probability of a failure
  - Also represented by q

Useful results

Read as "The probability of the first success after the xth trial"

$$P(X > x) = (1-p)^x$$

- This can also be interpreted as the "probability of failure on the first x trials"

Read as "The probability of the first success on one of the first x trials"

$$P(X \leq x) = 1 - (1-p)^x$$

- This can also be interpreted as "1 - probability of failure on x trials"

Measures of central tendency

- Mean of a Geometric Distribution:

$$E(x) = \mu = \frac{1}{p}$$

- The mode of all geometric distributions is 1

1.2. Examples

{S20-P51} Question 1:

The score when two fair six-sided dice are thrown is the sum of the two numbers on the upper faces.

a) Show that the probability that the score of 4 is $\frac{1}{12}$

The two dice are thrown repeatedly until a score of 4 is obtained. The number of throws taken is denoted by the random variable X.

b) Find the mean of X.

c) Find the probability that a score of 4 is first obtained on the 6th throw.

d) Find the probability of $x > 8$.

Solution:

Part (a) *To get a score of 4, there are three ways to do it:*

3+1

1+3

2+2

So now we calculate each probability and then add them up:

$$\left(\frac{1}{6}\right)\left(\frac{1}{6}\right) + \left(\frac{1}{6}\right)\left(\frac{1}{6}\right) + \left(\frac{1}{6}\right)\left(\frac{1}{6}\right) = \frac{1}{12}$$

Part (b)

This is a geometric distribution; its mean must be $\mu = \frac{1}{p}$

Because $p = \frac{1}{12}$

$$\therefore \mu = 12$$

Part (c)

First obtained in the 6th throw means $P(X = x)$ *, using the formula for geometric mean:*

$$P(X = 6) = \left(\frac{1}{12}\right)\left(1 - \frac{1}{12}\right)^{6-1}$$

$$P(X = x) \approx 0.0539$$

Part (d) To find $P(x > 8)$:

$$\left(1 - \frac{1}{12}\right)^8 = 0.498$$

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